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Levels of AI awareness and pedagogical skills among Nigerian teachers: The differentials of gender and teachers' experience

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A R T I C L E I N F O

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A B S T R A C T

The rapid diffusion of artificial intelligence (AI) across educational systems has intensified the need to assess teachers' readiness to engage with AI-informed pedagogy, particularly in underrepresented contexts. This study investigated the levels of AI awareness and AI pedagogical skills among secondary school teachers in Nigeria, with specific attention to gender and teaching experience differentials. A cross-sectional survey design was employed, drawing on a stratified random sample of 2,586 teachers across Nigeria's six geopolitical zones. AI awareness was measured using an eight-item binary scale, while AI pedagogical skills were assessed using the 20-item Educators' AI Literacy (EDAIL) scale ($\alpha = .977$). Data were analysed using descriptive statistics, independent samples t-tests, one-way ANOVA, correlation analysis, and ANCOVA. Findings indicate moderate levels of AI awareness ($M = 3.13$, $SD = 1.79$ on a 0–8 scale) but comparatively lower AI pedagogical skills ($M = 2.47$, $SD = 0.86$ on a 1–5 scale), suggesting a gap between conceptual familiarity and pedagogical application. Significant gender differences were observed, with male teachers reporting higher AI awareness ($d = 0.17$) and pedagogical skills ($d = 0.31$), although effect sizes were small. Teaching experience did not significantly predict AI awareness; however, early-career and mid-career teachers reported significantly higher AI pedagogical skills than experienced teachers ($\eta^2 = 0.008$). A moderate positive correlation ($r = .378$, $p < .001$) was found between AI awareness and pedagogical skills. These patterns remained robust after controlling for school type. These patterns may suggest that Nigerian secondary school teachers are navigating an early-to-intermediate phase of AI integration, conceptually familiar with AI technologies, yet not fully equipped to enact them pedagogically. The study contributes large-scale empirical evidence from Sub-Saharan Africa. It underscores the need for structured, equity-sensitive, and pedagogically grounded professional development frameworks to advance sustainable AI integration in education.

Introduction

The fifth industrial revolution is accelerating the integration of artificial intelligence (AI) across human endeavours, fundamentally reshaping global education and reconstructing teaching and learning processes both within and beyond formal school systems. The rapid advancement of AI technologies has established them as transformative tools with significant implications for how knowledge is designed, delivered, and assessed [1]. It makes it much easier for students and teachers to get information at an incredible speed. AI has changed how we design the curriculum, teach, and test students [2]. This includes activities outside of school. Without a doubt, AI seems to be having a big impact on the present and future of education systems around the world [3,4].

The rare disruption of AI in global educational systems has raised concerns about its potential impact on the Nigerian teaching profession [5,6]. The question is whether Nigerian teachers are fully aware of and ready for the rapid growth and development of artificial intelligence. The response to this is the main point of this paper. A typical Nigerian teacher is a dedicated and determined professional who has a big impact on the growth of future generations. They are very creative and flexible in how they teach, even when they face challenges [7]. Most of the

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teachers have teaching credentials, cultivating robust relationships with students via pertinent instruction and guidance [8], and many of them engage in continuous professional development to remain abreast of educational trends [9]. Awareness of Artificial Intelligence (AI) among Nigerian teachers has been documented in the literature, albeit with divergent findings [10,11] alluding to the conscious knowledge of teachers on the possibilities of integrating AI into education practices ranging from curriculum development, instructional delivery, assessment, and student supports [12,13]. In consonance with the UNESCO framework on AI [14], this paper conceptualised teachers' AI awareness as the understanding and perception of AI technologies and potential applications in education. It involves identifying how AI can support teaching tasks such as content development, lesson delivery, individualised learning, and automated assessment, among others. It entails the ability to demonstrate the foundational knowledge about AI technology, including how AI models work, the main categories of AI technologies with examples for each, and the ability to evaluate the suitability of specific AI tools for educational purposes while using validated tools [14]. Studies [15–17] shed light on various critical topics related to this understanding, describing the possibilities for beneficial transformation as well as the obstacles that presently hinder the effective implementation of AI.

Inadvertently, the post COVID-19 era is accentuating the demand for digital revolution in the educational systems, and AI-based teaching and learning is a veritable response to this transformation [18]. Alas, the desired transformation seems illusory within the Nigerian educational system due to the level of AI pedagogical skills and competencies among the teachers. For instance, Okunade [19], using a retrospective research approach, investigated how AI technologies enhance the general quality and efficacy of scientific teaching, but reported that teachers' competence is a barrier.

According to the UNESCO AI framework [14], AI pedagogical skills involve identifying and leveraging the pedagogical benefits of AI tools to facilitate subject-specific lesson planning, teaching, and assessment while mitigating the risks. Adeptly integrating AI into the design and facilitation of student-centred teaching practices to foster engagement, support differentiated learning, and enhance teacher–student interactions, with aims of promoting students' empathy, critical thinking, and problem-solving skills. In addition, AI pedagogical skills include the ability to critically assess AI's impact on teaching, learning, and assessment; plan and facilitate AI -immersed learning scenarios to support subject-specific or interdisciplinary learning, critical thinking, and problem-solving among students, leveraging data and feedback to continuously explore student-centred pedagogical innovation.

Gender differences in technology engagement are theoretically grounded in role congruity theory [20] and AI anxiety research [21], which suggest that socio-cultural stereotypes associating digital and STEM domains with masculinity may shape women's confidence, trust, and willingness to engage with emerging technologies such as AI. In the Nigerian context, where structural inequalities in access to digital infrastructure and technology-focused professional development persist [22], gender is therefore examined as a theoretically motivated moderator of AI awareness and pedagogical skills [23,24]. Similarly, teaching experience is theorised here not merely as tenure, but as a proxy for pedagogical routinisation and recency of training exposure. Drawing on TPACK theory [25] and evidence that pedagogical innovation may plateau without sustained professional renewal [26], experience is examined as a potential moderator, with the expectation that more recently trained teachers may demonstrate greater adaptability to AI-integrated instructional models. Hence, this paper investigated the Levels of AI awareness and pedagogical skills among Nigerian teachers, while focusing on the differentials of Gender and Teachers' Experience.

Problem statement

The integration of Artificial Intelligence (AI) in education is rapidly transforming teaching and learning processes worldwide [1,27]. In Nigeria, however, the adoption of AI-based educational technologies remains limited, particularly among teachers who are central to its implementation [28]. While AI offers promising tools to enhance pedagogical effectiveness, many Nigerian teachers seem to lack sufficient awareness and pedagogical skills to leverage this innovation and enhance learning in the classroom [29]. This gap is further complicated by demographic factors, among which gender and teaching experience are prominent, with the possibility of influencing teachers' levels of AI awareness and pedagogical skills. Despite growing global attention to AI readiness among educators, empirical evidence from Sub-Saharan Africa remains sparse, and critically, no large-scale study has examined how AI awareness translates into pedagogical competence among Nigerian secondary school teachers, or how this relationship is moderated by gender and teaching experience. This study addresses that gap directly, offering the first nationally representative, quantitative investigation of these constructs in the Nigerian context, grounded in the UNESCO AI competency framework and the extended TPACK model. The extended TPACK framework [25] and UNESCO AI competency model provide the theoretical scaffolding for this study, situating AI pedagogical skills within a broader conception of teacher digital competence. However, this study goes beyond applying these frameworks descriptively. By empirically testing whether AI awareness, a foundational construct in AI competency models, reliably predicts pedagogical competence at scale, the study subjects a core theoretical assumption to empirical scrutiny. The finding that awareness and pedagogical skills are meaningfully correlated yet empirically distinct ($r = .378$) contributes to refining TPACK-informed conceptions of AI readiness, suggesting that the pathway from technological knowledge to pedagogical enactment is neither automatic nor linear. This has implications for how AI competency frameworks are theorised, operationalised, and applied in teacher development contexts, particularly in under-resourced Sub-Saharan African settings where structural barriers mediate the awareness-to-practice transition. Hence, this study investigated the levels of AI awareness and pedagogical skills among Nigerian teachers, with a focus on examining the differentials of gender and teaching experience. The objectives include assessing the level of AI awareness and pedagogical skills among the Nigerian teachers; evaluating the role of gender and job experience in the level of AI awareness and pedagogical skills, and proposing strategic measures for improved levels of AI awareness and pedagogical skills.

Literature review

Artificial intelligence (AI) in education

The academic study of Artificial Intelligence (AI) in education has become one of the major arenas of academic research around the globe, and several empirical studies [30–32] outline its effect on education and learning practices in diverse educational systems. As a result, there are empirical examples in the literature that highlight the transformative nature of AI in the global education systems. On the example of the World Bank, Salimi [32] points out the willingness of this institution to analyse the power of AI to promote educational equity in sub-Saharan Africa. The article takes a universal approach to utilizing the emerging technologies to enhance education outcomes and freedom to develop sustainable and innovative initiatives in underdeveloped areas, using the example of AI. Similarly, Ifraheem et al. [31] ask questions about the radical impact of AI on the development of personalized learning experiences. They focus their research on the use of AI to achieve a shift in learning from teacher-centered to student-centered by making learning more participative and predictive. Researchers argue that DRC in the academic field needs to be implemented to create an interactive and individualized educational experience. However, [30] provides a systematic review of AI implementation in the educational field and notes that, despite the substantial potential of AI, many educators face challenges in understanding and using it effectively. The development of AI curricula for pre-tertiary education has also gained attention, with researchers exploring age-appropriate AI concepts and pedagogical approaches for younger learners [33]. Similarly, the integration of AI in vocational and technical education presents unique opportunities for developing sustainable workforce competencies [34]. They noted in their work that there is an immediate necessity in continuous research and professional growth to provide educators with the competencies they require to use AI in achieving better educational results.

AI awareness among teachers

The topography of AI consciousness among Nigerian educators is an indicator of a serious need for sophisticated educational systems that incorporate AI literacy. Despite the emerging indications of increasing awareness, there are steep hurdles that should be crossed to embrace AI successfully in classroom environments [35]. The study examines awareness of and use of smart learning environments in higher education institutions in Nigeria. The results reveal a striking lack of knowledge among instructors about AI applications, insufficient knowledge about AI technologies, as many students are owners of smartphones, and still, the teachers do not know much about AI technologies. The paper demonstrates the urgency of specific training and awareness programs that are focused on this shortcoming [36]. also show that a high level of awareness is not enough and constant interaction with the new technologies is essential. Their findings emphasize the need for prolonged professional growth [37]. Research the awareness of the AI of the teachers and note that the levels of awareness depend on socio-demographical factors, which require continuous training and exposure to AI tools to equip educators with these new technologies. According to [38], the level of education and awareness of AI among teachers is directly related; teachers with advanced education at higher education levels tend to have greater awareness and understanding of AI. Recent studies have extended this line of inquiry to pre-service teachers, examining their readiness to engage with AI technologies and the factors that influence their preparedness for AI-integrated classrooms [39]. Research has also explored the specific competencies required for teaching AI concepts at the middle school level, highlighting the need for developmentally appropriate pedagogical strategies [40]. This highlights the importance of providing professional development opportunities in the areas of AI literacy.

Pedagogical skills in the age of AI

The adoption of AI in education has stimulated an array of empirical studies [5,33,41–44] relating to the research on the way these technologies supplement teaching skills among instructors and transform the educational environment in general. The article by [41] focuses on assessing the training of the primary and middle school teachers in China on AI, revealing the structural features that promote the use of AI in instruction. Their conclusion suggests the holistic approach to embedding AI by combining pedagogical training, resources, and a helpful educational setting. To the same effect, [5] researched the effectiveness of AI-informed virtual simulations on enhancing responsive teaching abilities in pre-service teachers; the study of mixed-methods proves that AI-driven virtual environments provide educators with the opportunity to practice aspects of their instruction throughout their real-time performance, which enhances their responsive classroom dialogues. This study confirms the possibility of AI to improve the flexibility and responsiveness of pedagogy, which is crucial in modern educational institutions.

[42] presents the pedagogical trends determined by AI and explains how new technologies gradually alter the approach to teaching and classroom design. Their findings serve to emphasize the importance of having an effective theoretical grounding, as well as AI literacy, to attain successful educational practice and a balance between technological integration and pedagogical theory [44]. investigate the motivation aspects and didactic implications of applying AI into EFL classrooms and their results demonstrate positive effectiveness among students and teachers. The application of AI in teaching English to young learners presents unique pedagogical considerations, as educators must balance technological engagement with developmentally appropriate practices [45]. Systematic reviews of AI-enabled chatbots in education have synthesized evidence on both the benefits and implementation complexities of these tools across various educational contexts [46]. These reviews highlight the importance of pedagogical grounding in AI integration efforts. This can be explained by the larger story that AI will allow personalizing the learning process and improving the quality of teaching with the help of adaptive technologies [43]. researched how ChatGPT can be applied to strengthen the pedagogical competence of mathematics educators; the authors found that teachers can develop new lessons well, but cannot implement and evaluate teaching strategies. These results indicate a highly important area in which AI tools could streamline processes and improve the quality of teaching.

Beyond mathematics education, researchers have examined AI integration across diverse subject areas and educational levels. In English language education, studies have investigated how AI anxiety among language teachers affects their willingness to adopt AI-powered instructional tools [47]. This anxiety construct has been further explored among mathematics pre-service teachers, where it was found to significantly predict adoption intentions [48]. At the higher education level, faculty members at institutions such as Yarmouk University have reported generally positive attitudes toward AI integration while expressing concerns about institutional support and training adequacy [49]. International perspectives on AI integration in education reveal both common challenges and context-specific considerations. In Vietnamese secondary schools, researchers have identified infrastructural and pedagogical barriers that parallel those observed in Nigerian contexts, suggesting that developing countries face similar implementation hurdles [50]. Cross-national studies of digital literacy among university students have further underscored the foundational skills necessary for meaningful engagement with AI-enhanced learning environments [51]. The development of AI literacy among pre-service teachers has emerged as a critical priority, with researchers calling for systematic integration of AI competencies into teacher preparation curricula [33,52,53].

Inequalities between genders in AI awareness and pedagogical skills

Gender constructs have, over time, continued to creep into social science research, with the need to highlight many empirical studies concerning gender differences in AI consciousness and pedagogical skills. Empirical findings [21,23,54] indicate strong gender disparities in teachers' awareness of AI and in the acquisition of pedagogical skills. Russo et al. [21] inquire into the effects of AI-related anxiety on gender factors in the teaching practice and find that female teachers show more concern about AI than their male peers, thus determining willingness to implement AI technologies in the classroom. The intersection of AI and gender extends beyond teacher perceptions to include broader societal implications. Research has documented how AI systems themselves can perpetuate gender biases present in training data, raising important considerations for educators who must critically evaluate AI-generated content and recommendations [24]. These algorithmic biases have particular relevance for classroom applications where AI tools might inadvertently reinforce gender stereotypes [55]. It is critical to reduce AI anxiety, to strengthen the teacher's confidence and readiness to use AI tools, and hence improve instructional effectiveness [23]. are concerned with the trust of the teachers in the AI-based educational technology, and their results show that there is gender-specific differences in the trust levels of the teachers. Female educators note their unique fear of AI being implemented and controlled in schools, which explains why corporate development programmes should be customized to this fear. The assessment of readiness to integrate AI in both general and special education teachers in the UAE schools by Fteiha and colleagues [54] reveals significant gender differences; female teachers have more favourable views about AI, which implies increased awareness of its usefulness in education. However, even with positive intentions, the article reports that these dispositions do not always result in high AI usage among female instructors, which is an indication of the intricate nature of enthusiasm and structural restrictions.

The effect of teaching experience on AI awareness and pedagogical skills

Recent scholarship has increasingly examined the mediating role of teaching experience on technology-related competencies. In line with this, we expect to achieve major conclusions on the area of AI awareness and pedagogical proficiency among teachers in Nigeria [56]. Compare an analogy-based pedagogical approach to AI education among upper primary students and conclude that experienced teachers should show better implementation of the approach, which results in improved student engagement and conceptual understanding. It means that experience enhances adaptability and expands the innovative instructional knowledge, which is important to successful AI instruction. According to [57], there is a positive association between digital pedagogical competence and teaching performance among educators in public schools; the more experience an educator has using digital tools, including AI technologies, the better the expected learning outcomes [58]. are engaged in investigating teachers' awareness of pedagogical problems in teaching AI, and the findings suggest that experienced instructors not only identify these challenges but also develop effective strategies, thereby advancing their pedagogical competence [59]. investigates how pedagogical improvements are based on the experience of higher education teachers, who in their turn are more willing to use AI-based tools to enhance the learning process. This supports the conclusion that experience confers confidence and the ability to integrate AI effectively [60]. Underline the idea of incorporating Technological Pedagogical Content Knowledge (TPACK) to achieve success in the instruction and confirm that an experienced teacher has an inherent level of knowledge in terms of TPACK, which is required to be implemented in the curricula of the teacher to install digital technologies, including AI.

Methodology

Aim and objectives

The primary aim of this research is to explore and understand the current state of AI awareness and AI pedagogical skills among secondary school teachers in Nigeria, and to examine how these competencies vary across gender and teaching experience.

The specific objectives of this study are to:

1. Assess the level of AI awareness among secondary school teachers in Nigeria;
2. Evaluate the level of AI pedagogical skills among secondary school teachers in Nigeria;
3. Compare AI awareness between male and female teachers;
4. Compare AI pedagogical skills between male and female teachers;
5. Examine differences in AI awareness across early-career, mid-career, and experienced teachers; and
6. Examine differences in AI pedagogical skills across early-career, mid-career, and experienced teachers.

Research questions

To address these objectives, the Research Questions (RQs) were formulated:

RQ1. What is the current level of AI awareness among secondary school teachers?

RQ2. What is the current level of AI pedagogical skills among secondary school teachers?

RQ3. Are there significant gender differences in AI awareness and pedagogical skills?

RQ4. Are there significant experience-based differences in AI awareness and pedagogical skills?

Research hypotheses

The following null hypotheses were tested in this study:

$H1_o$: There is no significant difference in AI awareness scores between male and female secondary school teachers.
$H2_o$: There is no significant difference in AI pedagogical skills scores between male and female secondary school teachers.
$H3_o$: There are no significant differences in AI awareness scores among teachers with varying levels of teaching experience (early career, mid-career, and experienced).
$H4_o$: There are no significant differences in AI pedagogical skills scores among teachers with varying levels of teaching experience.

All hypotheses were tested at the $\alpha = .05$ significance level using appropriate parametric tests (independent samples t-tests for gender comparisons and one-way ANOVA for experience comparisons).

Method

Research design and data collection

This study employed a cross-sectional survey research design to investigate AI awareness and pedagogical skills among secondary school teachers in Nigeria. The cross-sectional approach was appropriate for capturing the current state of teachers' AI competencies and examining differences across gender and experience groups at a single point in time [61].

Data were collected between February, 2025 and September, 2025. Questionnaires were administered using Google Forms (online format). Teachers were assured of anonymity and confidentiality, and informed consent was obtained before participation. The questionnaire took approximately 15-20 minutes to complete.

Instrumentation

AI awareness measure

As seen in Table 1a, AI awareness was measured using eight binary

Table 1a
AI Awareness Scale Items (Binary Yes/No Response Format).

Item	Statement
1	Have you participated in any workshop or webinar on AI?
2	Have you heard about Artificial Intelligence (AI)?
3	Have you ever received training on AI?
4	Have you used any AI applications in your teaching practice?
5	Have you integrated AI into your lesson planning?
6	Do you have challenges in integrating AI tools into your teaching and assessment practices?
7	Have you received any training on the ethical implications of AI in education?
8	Have you at any time contributed to the development of any AI tools for educational purposes?

Note: Each item was scored dichotomously (Yes = 1, No = 0). A composite awareness score was computed by summing all eight items (range: 0–8). Higher scores indicate greater AI awareness.

items adapted from existing AI literacy frameworks [62]. The items assessed teachers' foundational knowledge and exposure to AI, including participation in AI workshops, training received, integration of AI in teaching practice, and awareness of ethical implications. Each item was scored dichotomously (Yes = 1, No = 0), and a composite awareness score was computed by summing all eight items, yielding a possible range of 0 to 8. Higher scores indicated greater AI awareness.

AI pedagogical skills scale (EDAIL)

The Educators' AI Literacy (EDAIL) scale was developed based on contemporary AI literacy frameworks [63] and the TPACK model [25] extended for AI. The scale consisted of 20 items measuring teachers' perceived competence in various aspects of AI-integrated pedagogy.

As detailed in Table 1b, items assessed competencies including understanding AI concepts, applying AI to teaching challenges, ethical considerations, creating inclusive AI-enhanced learning environments, and designing AI-driven assessments. Each item was rated on a 5-point Likert scale (1 = Very Poor, 2 = Below Average, 3 = Average, 4 = Above Average, 5 = Excellent). The scale demonstrated excellent internal consistency (Cronbach's α = 0.977). A composite pedagogical skills score was computed as the mean of all 20 items, ranging from 1 to 5.

Demographic variables

Gender was recorded as Male, Female, or Prefer not to say. Teaching experience was collected through open-ended responses and subsequently cleaned and categorized into three groups based on career stages: Early career (0-4 years), Mid-career (5-10 years), and Experienced (11+ years). School type was categorized as Federal, State, or Private.

Participants and sampling

The target population comprised secondary school teachers across Nigeria's six geopolitical zones (North-Central, North-East, North-West, South-East, South-South, and South-West). A stratified random sampling technique was employed to ensure proportional representation across zones. The sampling frame was obtained from State Ministries of Education records, which provided comprehensive lists of registered secondary schools and their teaching staff.

Stratification was first applied at the zonal level, with the sample size for each zone proportional to its teacher population. Within each zone, schools were randomly selected using a computer-generated random number sequence. Subsequently, teachers within selected schools were randomly sampled from staff lists. This multi-stage random sampling approach minimized selection bias and enhanced the representativeness of the sample.

Participants accessed the questionnaire online, and 2,586 valid responses were received, yielding a response rate of 80.8%. This sample size substantially exceeds the minimum requirements for the planned statistical analyses (t-tests, ANOVA, ANCOVA) as determined by a priori power analysis [64], which indicated that a sample of approximately 400 would be sufficient to detect small-to-medium effects (Cohen's d = 0.3) with 80% power at α = .05.

Table 2 presents the demographic characteristics of the sample. As shown, the sample consisted of 1,701 males (65.8%), 881 females (34.1%), and 4 participants (0.2%) who preferred not to disclose their gender. Participants were drawn from State (76.1%), Federal (16.5%), and Private (7.4%) schools. Regarding teaching experience, 10.4% were early career (0-4 years), 31.1% were mid-career (5-10 years), 54.0% were experienced (11+ years), and 4.6% had unknown experience after data cleaning.

Data analysis

To ensure rigor, transparency, and structure, data were analyzed using Python 3.11 with pandas, numpy, scipy, statsmodels, and pingouin libraries. Before analysis, data cleaning procedures were systematically applied. Missing data were minimal: only 31 values were missing across the entire dataset of 2,586 responses, confined to two non-focal variables (Subject Taught, n = 30; Teaching Role/Position, n = 1). These missing values were not imputed, as they pertained to demographic descriptors rather than outcome variables, and listwise deletion was applied where relevant in subgroup analyses. The teaching experience variable required substantial cleaning due to inconsistent response formats (e.g., "Ten years," "7 years," "6 YEARS"). A standardised text-parsing procedure was applied to extract numeric values, successfully converting 2,468 of 2,586 responses. The remaining 118 responses that could not be reliably converted, including non-numeric entries such as "Good teaching experience", were categorised as Unknown and excluded from experience-based analyses. Outlier screening of the teaching experience variable identified implausible values (e.g., entries exceeding reasonable career lengths), which were absorbed into the Experienced (11+ years) category through the three-tier categorical classification, thereby minimising their influence on group comparisons. No missing data were present on the primary outcome variables (AI Awareness Score and AI Pedagogical Skills Score), as both were computed as composite scores from fully completed scale items. Descriptive statistics (frequencies, percentages, means, standard deviations) were then computed for all variables. For hypothesis testing, independent samples t-tests (Welch's) were conducted to compare AI awareness and pedagogical skills between male and female teachers. One-way analysis of variance (ANOVA) was used to examine differences across the three experience categories, with Tukey HSD post-hoc tests

Table 1b
Descriptive statistics for AI awareness and pedagogical skills (N = 2,586).

Item	Description	Mean	SD
1	Understanding of Artificial Intelligence (AI)	2.48	0.98
2	Ability to apply AI concepts to solve teaching challenges	2.44	0.97
3	Knowledge of ethical considerations in AI use	2.52	1.02
4	Ability to explain how AI systems are trained and tested	2.38	0.99
5	Knowledge of different AI technologies in education	2.45	0.99
6	Method of incorporating AI tools into teaching practices	2.46	0.98
7	Ability to customize AI tools for specific teaching needs	2.40	0.98
8	Ability to create AI tools for educational challenges	2.35	0.99
9	Understanding of inclusive education principles	2.74	1.04
10	Ability to use AI for inclusive learning environments	2.50	1.01
11	Ability to adapt AI resources for diverse student needs	2.52	1.00
12	Ability to find AI tools for subject-specific lessons	2.50	0.99
13	Skill at leveraging AI for instructional design	2.46	0.98
14	Confidence integrating AI into student-centered teaching	2.48	0.99
15	Ability to design AI-enhanced formative assessments	2.44	0.98
16	Skill in evaluating ethical and educational impacts of AI	2.47	0.99
17	Skill in generating AI-driven content	2.47	1.00
18	Skill in evaluating AI tools for learning analytics	2.43	0.99
19	Skill in designing AI tools for cross-subject learning	2.40	0.98
20	Ability to use AI for data-driven teaching adjustments	2.44	0.99

Note: All items measured on a 5-point scale (1=Very Poor to 5=Excellent). Cronbach's α = 0.977.

Table 2
Sample characteristics (N = 2,586).

Characteristic	Category	Frequency	Percentage
Gender	Male	1,701	65.8%
	Female	881	34.1%
	Prefer not to say	4	0.2%
School Type	State	1,968	76.1%
	Federal	426	16.5%
	Private	192	7.4%
Teaching Experience	Early career (0-4 years)	269	10.4%
	Mid-career (5-10 years)	803	31.1%
	Experienced (11+ years)	1,396	54.0%
	Unknown	118	4.6%

applied when significant main effects were found. Analysis of covariance (ANCOVA) was conducted to control for potential confounding effects of school type.

Effect sizes were calculated for all significant findings: Cohen's d for t-tests and η^2 for ANOVA. Assumptions of normality and homogeneity of variance were checked using Shapiro-Wilk tests and Levene's tests, respectively. All statistical tests were two-tailed with an alpha level set at .05.

Ethical considerations

Informed consent was secured from all participants, ensuring they understood the study's purpose and voluntary nature. Data were anonymized to protect participants' identities, and findings were reported in a manner that respected privacy, thus aligning with GDPR principles.

Ensuring replicability and trustworthiness

To enhance replicability, numerous supplementary methodological measures were implemented. The study adhered to open science standards by meticulously describing all data processing and analytical techniques. The comprehensive Python analysis script, encompassing all data cleaning procedures, variable transformations, and statistical analyses, including composite score construction, Welch's t-tests, one-way ANOVA with Tukey HSD post-hoc tests, correlation analysis, and ANCOVA, is available upon request from the corresponding author. Researchers seeking to replicate or extend these analyses are encouraged to contact the authors directly. Future iterations of this work will aim to deposit the anonymised dataset and full analysis script in an open repository such as the Open Science Framework (OSF) or Zenodo, consistent with emerging open science standards in educational technology research. This allows other researchers to replicate the analyses precisely as performed. The complete instrument (EDAIL scale), comprising all 20 items with their precise phrasing, is presented in Table 1, enabling other researchers to implement the same measure in their respective situations. The response format and grading methodologies are clearly delineated, enabling cross-study comparisons.

The stratified random sample method was thoroughly documented to facilitate replication in other Nigerian states or Sub-Saharan African nations. The source of the sampling frame, the criteria for stratification, and the techniques for random selection are all specified.

Fourth, all statistical assumptions were evaluated and documented, encompassing normality assessments, homogeneity of variance tests, and outlier analysis. In instances of assumption violation, suitable robust procedures, such as Welch's t-test, were utilised and recorded.

Fifth, effect estimates accompanied by confidence intervals are presented for all significant results, facilitating meta-analytic comparisons in further research. This method enables readers to evaluate the practical importance of discoveries in addition to their statistical significance.

The dependability of the EDAIL scale was meticulously evaluated using Cronbach's alpha, with item-level statistics presented to provide transparency in scale performance. The study's limitations are fully addressed, offering readers a clear comprehension of the parameters within which findings may be interpreted and duplicated.

Limitations

This study provides significant insights into AI knowledge and pedagogical competencies among Nigerian secondary school educators, although it possesses some drawbacks. The cross-sectional design precludes causal inferences, and dependence on self-reports may induce bias. The generalisability is limited due to the emphasis on secondary school educators and the minimal representation of private school participants. Crucial contextual elements, including access to technology, previous training, and educational resources, were not assessed, and the classification of teaching experience may have overly simplified variability. The lack of qualitative data constrains comprehension of the fundamental causes behind reported disparities in gender and experience. The findings represent a 2025 snapshot inside a swiftly changing AI ecosystem, and while the EDAIL scale is dependable, it necessitates additional validation across varied scenarios. Notwithstanding these constraints, the study offers a substantial empirical contribution to teacher AI readiness in Sub-Saharan Africa, documents demographic disparities, and establishes a foundation for subsequent research.

Results

Descriptive statistics

Table 3 above presents descriptive statistics for the main study variables. The mean AI awareness score was 3.13 (SD = 1.79) on a 0-8 scale, indicating moderate levels of AI awareness among teachers. The mean AI pedagogical skills score was 2.47 (SD = 0.86) on a 1-5 scale, suggesting that teachers rated their AI teaching competencies slightly below the "average" midpoint.

Fig. 1 displays the distributions of both variables. AI awareness scores showed a slight positive skew, with most teachers scoring between 2 and 4. AI pedagogical skills scores were approximately normally distributed, with a mean of 2.5, and fewer teachers rated themselves at the extremes.

Gender differences in AI awareness and pedagogical skills

Table 4 presents the results of independent samples t-tests comparing male and female teachers. For AI awareness, a significant difference was found ($t(2580) = 4.248$, $p < 0.001$), with male teachers (M = 3.24, SD = 1.84) reporting higher awareness than female teachers (M = 2.93, SD = 1.66). The effect size was small (Cohen's d = 0.17, 95% CI [0.10, 0.24]).

For AI pedagogical skills, a significant difference was also observed ($t(2580) = 7.698$, $p < 0.001$), with male teachers (M = 2.56, SD = 0.87) reporting higher skills than female teachers (M = 2.29, SD = 0.81). The effect size was small-to-moderate (Cohen's d = 0.31, 95% CI [0.23, 0.39]).

Fig. 2 visualizes these gender differences through box plots, illustrating the higher central tendency and wider distribution among male teachers for both outcomes.

Experience differences in AI awareness and pedagogical skills

Table 5 summarizes experience-based differences. For AI awareness, one-way ANOVA revealed no significant differences across experience categories ($F(2, 2465) = 2.967$, $p = 0.052$, $\eta^2 = 0.002$). Although the p-value approached significance, post-hoc comparisons were not warranted given the non-significant omnibus test.

Post-hoc Comparisons for AI Pedagogical Skills:

Comparison	Mean Difference	p	95% CI	Significance
Early career vs. Experienced	0.194	0.002	[0.060, 0.328]	*
Mid-career vs. Experienced	0.141	< 0.001	[0.052, 0.230]	*
Early career vs. Mid-career	0.053	0.663	[-0.089, 0.195]	ns

For AI pedagogical skills, a significant main effect of experience was found ($F(2, 2465) = 10.201$, $p < 0.001$, $\eta^2 = 0.008$), though the effect size was small. Tukey HSD post-hoc comparisons revealed that early career teachers (M = 2.60, SD = 0.95) scored significantly higher than experienced teachers (M = 2.41, SD = 0.82), with a mean difference of 0.194 (p = 0.002, 95% CI [0.060, 0.328]). Mid-career teachers (M =

Table 3
Descriptive statistics for AI awareness and pedagogical skills (N = 2,586).

Variable	N	Mean	SD	Median	Min	Max	Skewness	Kurtosis	Cronbach's α
AI Awareness (0-8 scale)	2,586	3.13	1.79	3.0	0	8	0.42	-0.28	–
AI Pedagogical Skills (1-5 scale)	2,586	2.47	0.86	2.4	1	5	0.51	-0.15	0.977

Note: AI Awareness was measured using 8 binary items (Yes=1, No=0). AI Pedagogical Skills was measured using 20 self-evaluation items on a 5-point scale (1=Very Poor, 2=Below Average, 3=Average, 4=Above Average, 5=Excellent).

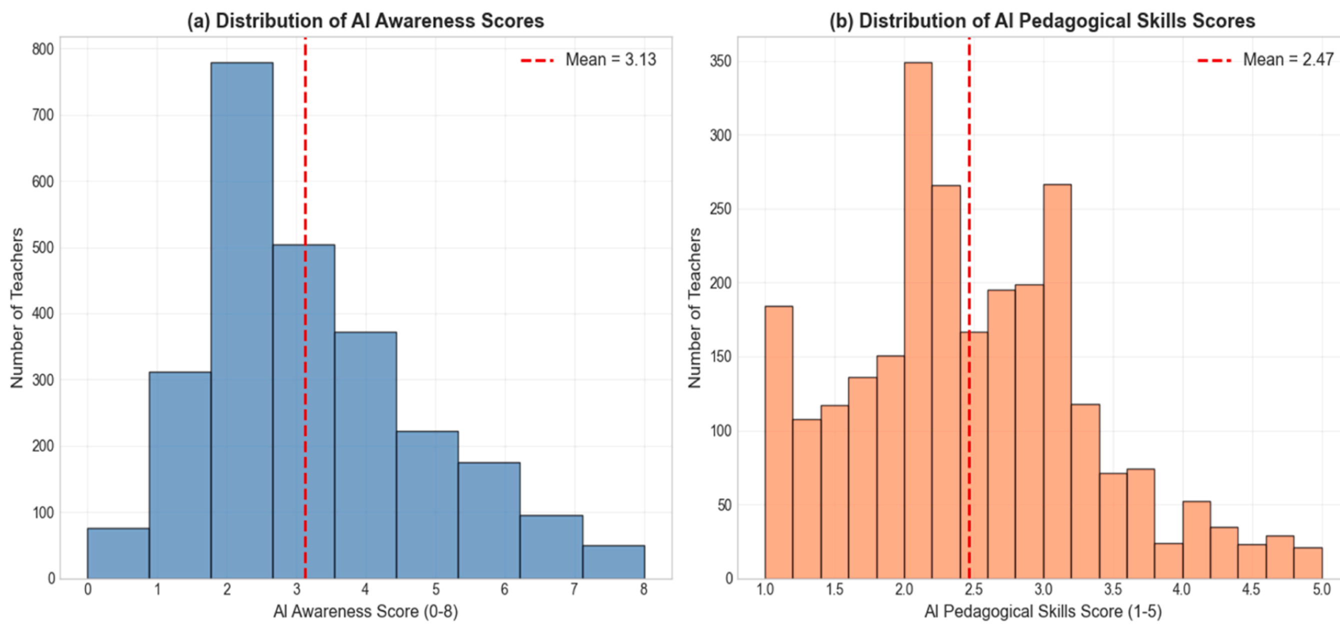

Fig. 1. Distribution of (a) AI Awareness Scores and (b) AI Pedagogical Skills Scores.

Table 4
Gender differences in AI awareness and pedagogical skills.

Variable	Gender	N	Mean	SD	t	Df	p	Cohen's d	95% CI
AI Awareness	Male	1,701	3.24	1.84	4.248	2,580	< 0.001	0.17	[0.10, 0.24]
	Female	881	2.93	1.66					
AI Pedagogical Skills	Male	1,701	2.56	0.87	7.698	2,580	< 0.001	0.31	[0.23, 0.39]
	Female	881	2.29	0.81					

Note: Independent samples t-tests (Welch's) were conducted. Participants who preferred not to disclose gender (n=4) were excluded from this analysis. Effect size interpretation: d = 0.2 (small), 0.5 (medium), 0.8 (large).

2.55, SD = 0.89) also scored significantly higher than experienced teachers, with a mean difference of 0.141 ($p < 0.001$, 95% CI [0.052, 0.230]). No significant difference was found between early-career and mid-career teachers ($p = 0.663$).

Fig.s 3 and 4 provide visual representations of these experience differences. Fig. 3 displays mean scores with error bars, while Fig. 4 presents box plots showing the full distribution of pedagogical skills scores across experience groups.

Relationship between AI awareness and pedagogical skills

A significant positive correlation was found between AI awareness and AI pedagogical skills ($r = 0.378$, $p < 0.001$), indicating that teachers with higher awareness tended to report higher pedagogical skills. This represents a moderate effect size according to Cohen's (1988) conventions.

Fig. 5. Scatter Plot Showing Relationship Between AI Awareness and Pedagogical Skills

Controlling for school type

To determine whether the observed gender and experience differences remained robust after accounting for school type, ANCOVA was conducted with school type (Federal, State, Private) as a covariate. As shown in Table 6, the effects of gender on both AI awareness ($F(1, 2459) = 38.79$, $p < 0.001$, $\eta^2 = 0.016$) and AI pedagogical skills ($F(1, 2459) = 51.89$, $p < 0.001$, $\eta^2 = 0.021$) remained significant after controlling for school type. The effect of experience on AI pedagogical skills also remained significant ($F(2, 2459) = 10.90$, $p < 0.001$, $\eta^2 = 0.009$). School type itself was not a significant predictor of either outcome ($p > 0.05$), suggesting that the observed patterns were consistent across Federal, State, and Private schools.

Summary of hypothesis testing

Table 7 provides a comprehensive summary of all hypothesis tests. Hypotheses H1, H2, and H4 were supported, indicating significant gender differences in both AI outcomes and significant experience differences in pedagogical skills. Hypothesis H3 was not supported, as no

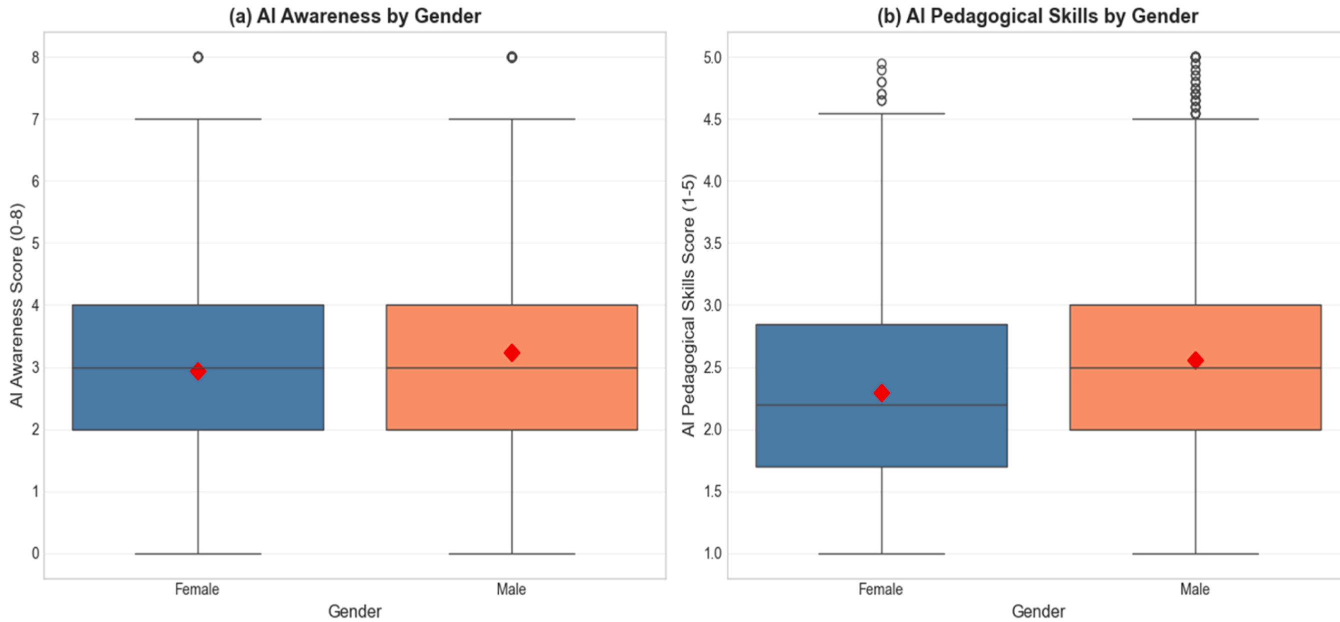

Fig. 2. Gender Differences in (a) AI Awareness and (b) AI Pedagogical Skills.

Table 5
Experience differences in AI awareness and pedagogical skills.

Variable	Experience Category	N	Mean	SD	ANOVA	Post-hoc (Tukey HSD)
AI Awareness	Early career (0-4 years)	269	3.12	1.77	$F(2, 2465) = 2.967$, $p = 0.052$, $\eta^2 = 0.002$	No significant pairwise differences
	Mid-career (5-10 years)	803	3.26	1.83		
	Experienced (11+ years)	1,396	3.07	1.77		
AI Pedagogical Skills	Early career (0-4 years)	269	2.60	0.95	$F(2, 2465) = 10.201$, $p < 0.001$, $\eta^2 = 0.008$	Early > Experienced* ($p = 0.002$)
	Mid-career (5-10 years)	803	2.55	0.89		Mid > Experienced* ($p < 0.001$)
	Experienced (11+ years)	1,396	2.41	0.82		Early vs. Mid ($p = 0.663$)

Note: Participants with unknown experience (n=118) were excluded from this analysis. Effect size η^2 interpretation: 0.01 (small), 0.06 (medium), 0.14 (large). *Significant at $p < 0.05$*

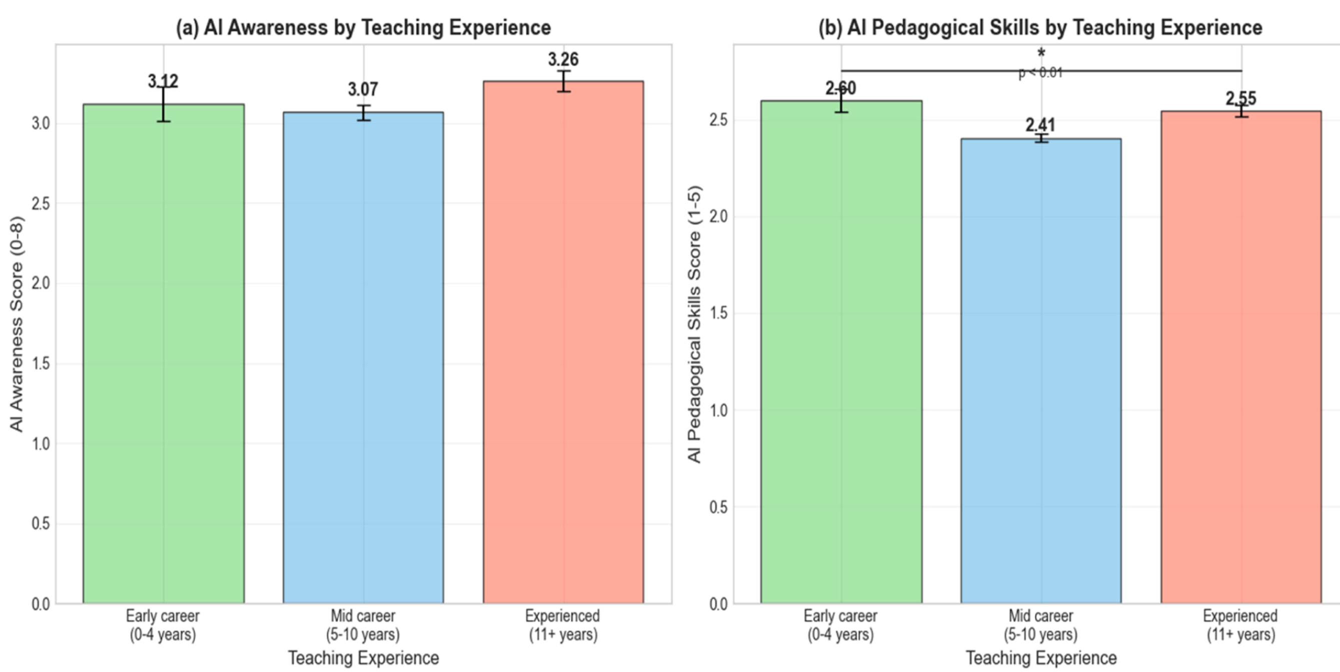

Fig. 3. Experience Differences in (a) AI Awareness and (b) AI Pedagogical Skills.

significant experience differences were found in AI awareness.

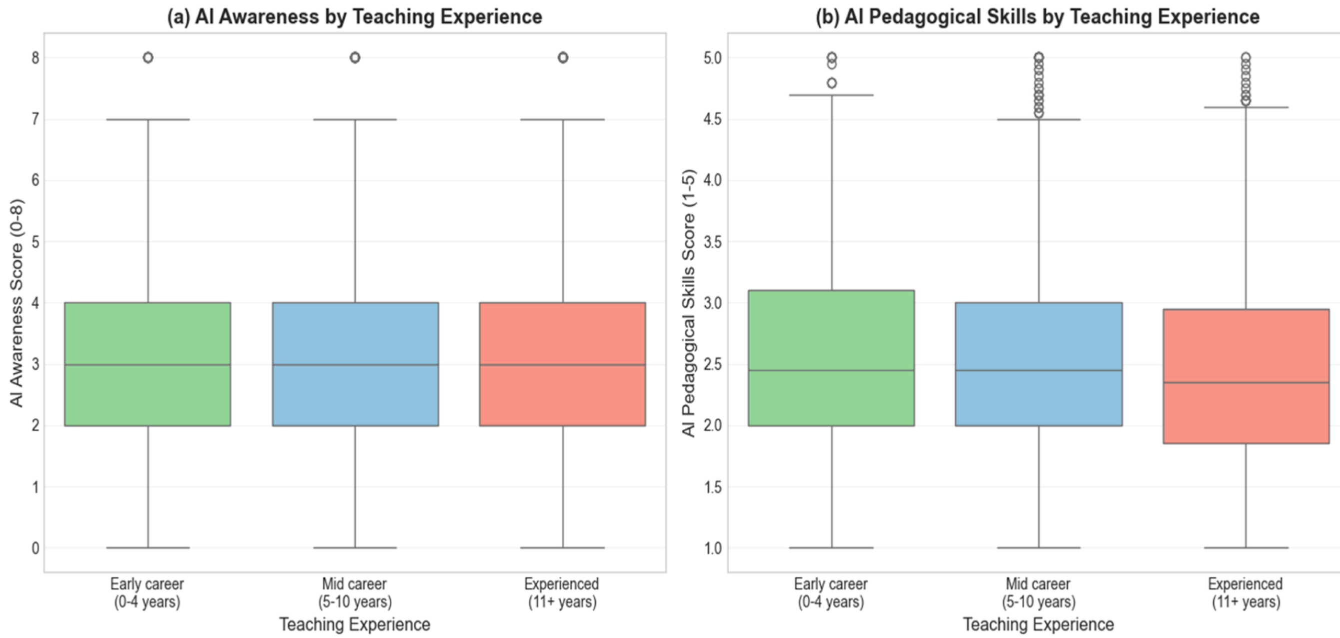

Fig. 4. Box Plots of AI Pedagogical Skills by Experience Category.

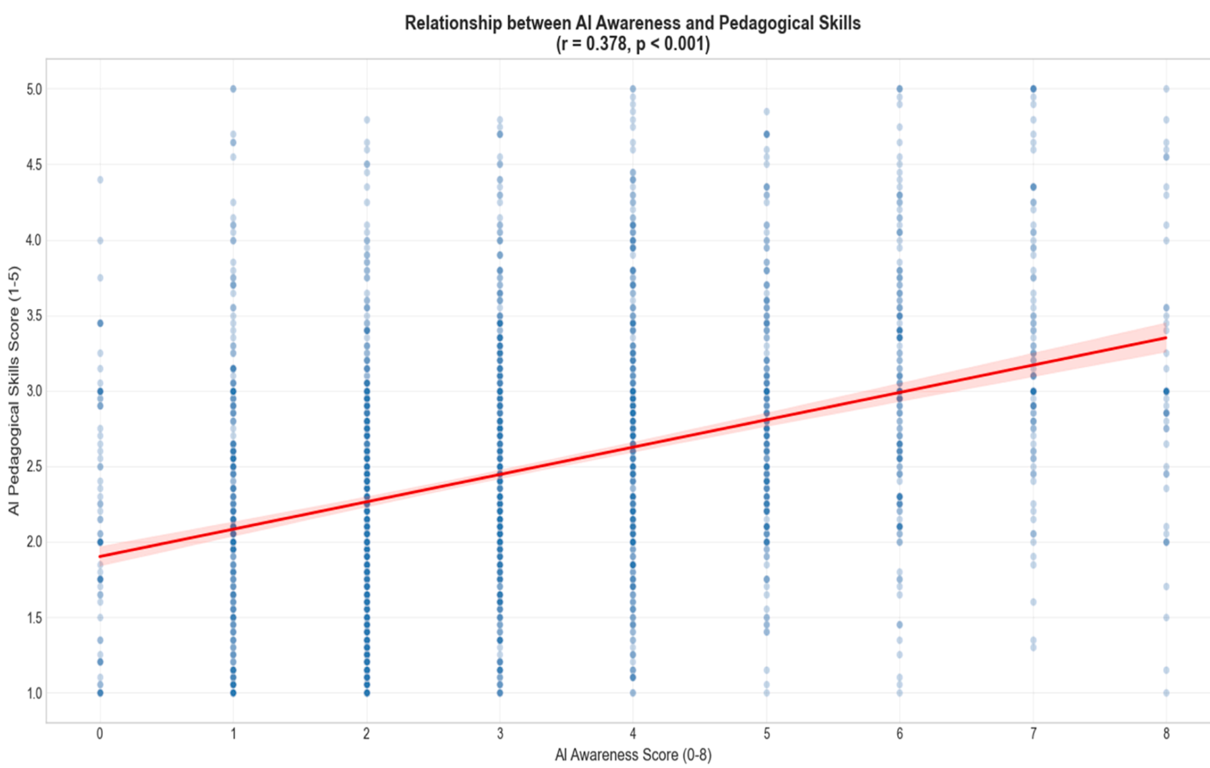

Fig. 5. illustrates this relationship through a scatter plot with a fitted regression line, demonstrating the positive association between the two constructs.

Discussion

This study examined the levels of AI awareness and AI pedagogical skills among Nigerian secondary school teachers, with particular attention to gender and teaching experience differentials. The findings contribute to the growing discourse on teacher AI readiness in Sub-Saharan Africa and align with the broader scholarship on artificial intelligence integration in education.

First, the results indicate that Nigerian secondary school teachers demonstrate moderate levels of AI awareness (M = 3.13 on a 0–8 scale) but relatively low AI pedagogical skills (M = 2.47 on a 1–5 scale). This discrepancy between awareness and applied pedagogical competence is theoretically and empirically consistent with prior studies. While awareness of AI technologies is increasingly reported among educators, the translation of this awareness into classroom-level integration remains uneven [30,37]. The moderate awareness scores observed in this study suggest that teachers possess foundational exposure to AI concepts, workshops, and discourse, yet the below-average pedagogical

Table 6
ANCOVA results: Effects of gender and experience controlling for school type.

Dependent Variable	Source	Sum of Squares	df	Mean Square	F	p	η^2
AI Awareness	Gender	27.88	1	27.88	38.79	< 0.001	0.016
	Experience	3.38	2	1.69	2.35	0.095	0.002
	School Type	2.92	2	1.46	2.03	0.131	0.002
	Residual	1,767.41	2,459	0.72			
AI Pedagogical Skills	Gender	37.30	1	37.30	51.89	< 0.001	0.021
	Experience	15.66	2	7.83	10.90	< 0.001	0.009
	School Type	3.02	2	1.51	2.10	0.123	0.002
	Residual	1,767.41	2,459	0.72			

Note: School type (Federal, State, Private) was entered as a categorical covariate. Sample size for ANCOVA: n = 2,465 after excluding cases with missing data

Table 7
Summary of hypothesis testing results.

Hypothesis	Independent Variable	Dependent Variable	Test	Test Statistic	p	Effect Size	Decision
H1	Gender	AI Awareness	t-test	t(2580) = 4.248	< 0.001	d = 0.17	**Reject null**
H2	Gender	AI Pedagogical Skills	t-test	t(2580) = 7.698	< 0.001	d = 0.31	**Reject null**
H3	Experience	AI Awareness	ANOVA	F(2, 2465) = 2.967	0.052	$\eta^2 = 0.002$	Fail to reject
H4	Experience	AI Pedagogical Skills	ANOVA	F(2, 2465) = 10.201	< 0.001	$\eta^2 = 0.008$	**Reject null**

self-ratings imply limited confidence in designing AI-enhanced instruction, assessment, and differentiated learning experiences. This pattern resonates with findings by [43], who reported that teachers may be able to conceptualize AI-supported lessons but struggle with implementation and evaluative dimensions of AI integration.

From a competence-framework perspective, the findings echo the argument advanced by UNESCO [14] that AI literacy for teachers must extend beyond conceptual understanding to include pedagogical design, ethical reasoning, and data-informed instructional decision-making. The strong internal consistency of the EDAIL scale ($\alpha = .977$) further suggests that the construct of AI pedagogical skills, as operationalized in this study, captures a coherent domain aligned with extended TPACK conceptualizations [25,60]. Thus, the observed skill gap underscores the urgency of structured professional development that explicitly links AI tools to pedagogical theory and classroom practice, rather than treating AI as a purely technical competency.

Second, significant gender differences were found in both AI awareness and AI pedagogical skills, with male teachers reporting higher scores than female teachers. Although the effect sizes were small (d = 0.17 and d = 0.31), their practical significance warrants careful interpretation. A Cohen's d of 0.31 for pedagogical skills, while statistically small, may carry practical relevance at the population level. If sustained across the broader teaching workforce, even modest average differences in AI pedagogical confidence could contribute to disparities in classroom practice and student exposure to AI-enhanced learning, though this inference requires longitudinal confirmation. This is particularly consequential in a context where female teachers constitute a substantial proportion of the workforce yet may be systematically less equipped to leverage AI tools, not due to capability deficits, but due to structural and socio-cultural barriers to training and confidence-building. These findings are consistent with role congruity theory [20], which predicts that in contexts where digital and STEM domains are implicitly masculinised, women may experience greater AI anxiety, lower technology trust, and reduced confidence in adopting emerging tools [23,21]. In the Nigerian context specifically, structural factors, including unequal access to digital infrastructure and technology-focused professional networks, may compound these socio-cultural dynamics, producing the modest but consistent gender gap observed here [24]. However, the present findings contrast with research suggesting that female educators may exhibit more favourable attitudes toward AI or equal readiness under supportive training conditions [54]. This divergence foregrounds an important equity consideration: the observed gender gap is unlikely to reflect inherent differences in capability, but rather the unequal distribution of structural enablers, reliable electricity, internet connectivity, device access, and technology-focused professional development that disproportionately constrain female teachers' opportunities to engage with AI tools in the Nigerian context [28,65]. Addressing this gap is therefore not merely a matter of individual capacity-building but of systemic equity investment. Sustainable AI integration in Nigerian education will require deliberate attention to the structural conditions that either enable or foreclose teachers' meaningful engagement with AI, particularly for women, early-career teachers in under-resourced schools, and educators in geopolitically marginalised zones. Without addressing these foundational inequities, professional development initiatives risk reproducing existing disparities rather than closing them. This divergence may be context-sensitive. In Nigeria, structural factors such as differential access to advanced digital infrastructure, informal professional networks, or technology-focused professional development could mediate gendered outcomes. Importantly, the small magnitude of the effects suggests that while statistically significant, gender differences should not be overstated; rather, they highlight the need for inclusive AI capacity-building initiatives that address confidence, anxiety, and structural barriers without reinforcing deficit narratives.

Third, teaching experience did not significantly predict AI awareness but did predict AI pedagogical skills, albeit with small effect sizes. Interestingly, early-career and mid-career teachers reported significantly higher pedagogical skills than experienced teachers. This pattern challenges the assumption that longer tenure necessarily confers greater digital or AI-related competence. While the effect size is small ($\eta^2 = 0.008$, accounting for less than 1% of variance), its practical meaning warrants attention: experienced teachers, who constitute 54% of this sample, report lower AI pedagogical confidence than their less experienced colleagues. Given that this group delivers the majority of classroom instruction in Nigerian secondary schools, this pattern may have implications for AI integration at scale, though the small effect size cautions against overstating the magnitude of this difference. While prior studies have emphasized the positive relationship between experience and instructional adaptability [56,59], more recent evidence suggests that digital pedagogical innovation may be more strongly associated with recency of training and exposure to emerging technologies than with cumulative years of service.

The absence of significant experience differences in AI awareness indicates that exposure to AI discourse may be relatively diffuse across career stages. However, the lower pedagogical skill ratings among experienced teachers may reflect entrenched pedagogical routines, limited re-training opportunities, or lower engagement with emerging AI-driven instructional models. This interpretation aligns with research showing that while teaching quality can increase with experience, the trajectory is not linear and may plateau without sustained professional renewal [60]. Consequently, AI-focused professional development in Nigeria may need to be differentiated, with targeted support for experienced teachers to facilitate pedagogical retooling.

The moderate positive correlation between AI awareness and AI pedagogical skills (r = .378) reinforces the conceptual linkage between knowledge and practice. Teachers who are more aware of AI technologies are more likely to perceive themselves as capable of integrating them pedagogically. This finding is broadly consistent with structural models of AI adoption in education [41] and suggests that awareness-building initiatives may serve as foundational levers for competence development, though the moderate correlation (r = .378) indicates that awareness accounts for only a modest proportion of variance in pedagogical skills, and other factors, including training quality, infrastructure access, and institutional support, are likely to play important complementary roles. Nevertheless, the correlation is not strong enough to imply equivalence; awareness alone is insufficient for robust pedagogical transformation. This underscores the need for experiential, practice-oriented AI training, including simulations, design-based learning, and collaborative inquiry, as advocated in recent AI pedagogy research [58].

Importantly, the robustness of gender and experience effects after controlling for school type suggests that the observed patterns are not merely artifacts of institutional variation between Federal, State, and Private schools. This strengthens the internal validity of the findings and indicates that demographic variables, rather than institutional classification, play a more salient role in shaping AI readiness within this sample. Collectively, the findings suggest that Nigerian secondary school teachers may be navigating an early-to-intermediate phase of AI integration: broadly aware of AI technologies, yet not yet fully equipped to translate that awareness into confident, structured pedagogical practice. In line with global calls for systematic AI competence frameworks for educators [66,67], the study underscores the necessity of structured, equity-sensitive, and pedagogically grounded AI professional development. Critically, however, the sustainability of AI integration in Nigerian secondary education cannot rest on teacher competence alone. Infrastructure deficits, including unreliable power supply, limited broadband access, and insufficient device availability, represent foundational barriers that professional development cannot overcome in isolation [50]. For AI integration to be sustainable and equitable in the African context, coordinated investment across three interdependent levels is required: individual teacher competence, institutional support structures, and national digital infrastructure. Framing AI readiness purely as a teacher-level problem risks misattributing systemic failures to individual shortcomings, and in doing so, may inadvertently deepen rather than address educational inequities across Nigeria's diverse geopolitical zones.

Conclusion

This study investigated the levels of artificial intelligence (AI) awareness and AI pedagogical skills among Nigerian secondary school teachers, with particular attention to gender and teaching experience differentials. Drawing on a large, stratified national sample, the findings provided a comprehensive empirical snapshot of teacher AI readiness in Sub-Saharan Africa to date. In doing so, the study responds to growing global calls for systematic evidence on educators' preparedness to engage meaningfully with AI-driven transformations in teaching and learning, as emphasized in recent AI competence frameworks for educators [14,66,67].

The results indicate that while Nigerian secondary school teachers demonstrate moderate levels of AI awareness, their self-reported AI pedagogical skills remain below the "average" benchmark. This gap between conceptual familiarity and pedagogical enactment highlights a critical developmental threshold. Teachers appear to be aware of AI technologies and their potential applications, yet many lack the confidence and structured competence required to design, implement, and critically evaluate AI-enhanced instructional practices. In line with international research on AI integration in education [30,43], the findings reinforce the argument that awareness alone is insufficient to achieve pedagogical transformation. Sustainable AI integration requires targeted professional development that bridges knowledge, ethics, instructional design, and classroom application.

The study further reveals statistically significant gender differences in both AI awareness and AI pedagogical skills, with male teachers reporting higher scores. Although effect sizes were small, the consistency of the pattern suggests that gender remains a relevant structural variable in AI readiness. These findings must be interpreted cautiously to avoid deficit framings; however, they resonate with broader scholarship on AI-related anxiety, trust, and gendered perceptions of technology [21,23]. The implication is not that female teachers lack capability, but that contextual, socio-cultural, and confidence-related factors may shape differential engagement with emerging technologies. Equity-sensitive AI capacity-building initiatives are therefore essential to ensure inclusive digital transformation within the teaching workforce.

Regarding teaching experience, the absence of significant differences in AI awareness suggests that exposure to AI discourse is relatively distributed across career stages. However, early-career and mid-career teachers reported higher AI pedagogical skills than experienced teachers. These findings challenge linear assumptions about experience and instructional competence and align with evidence that technological adaptability may be more strongly associated with recency of training than with cumulative years of service [26]. The results underscore the importance of continuous professional renewal and differentiated professional development models that specifically support experienced teachers in retooling pedagogical practices for AI-enhanced environments.

The moderate positive correlation between AI awareness and AI pedagogical skills confirms that knowledge and practice are meaningfully related, yet distinct. Enhancing foundational AI literacy is likely to contribute to pedagogical confidence, but structured, practice-oriented interventions remain indispensable. From a policy and systems perspective, the findings carry clear implications for educational equity and the sustainable integration of AI in Nigerian education. National and sub-national education authorities must move beyond isolated workshops toward sustained, evidence-based AI professional development ecosystems that are explicitly designed to reach underserved populations, including female teachers, experienced teachers resistant to retooling, and educators in zones with limited digital infrastructure. Equity-sensitive AI integration demands not only training investment but parallel infrastructure development: stable electricity, affordable internet access, and school-level device provision are preconditions, not afterthoughts, for meaningful AI adoption. In the African context more broadly, these structural realities must be centred in policy design rather than relegated to limited sections of research papers [32,50]. The present findings, situated within Nigeria's six geopolitical zones, offer a contextually grounded evidence base for such policy conversations across Sub-Saharan Africa. Such ecosystems should integrate ethical reasoning, instructional design principles, inclusive education considerations, and data-informed teaching practices, consistent with contemporary AI competency frameworks [14,66,68].

This study contributes to knowledge in three distinct ways, each with implications for theory, methodology, and practice. Theoretically, the study advances TPACK scholarship and AI competency frameworks by providing large-scale empirical evidence that AI awareness and AI pedagogical skills, while meaningfully correlated (r = .378), constitute empirically distinct constructs. This finding challenges a linear assumption embedded in many AI readiness frameworks, namely, that foundational awareness naturally progresses into pedagogical competence, and instead points to a structural gap between knowing about AI and being able to enact it instructionally. In contexts such as Nigeria, where infrastructure deficits, limited professional development, and curriculum rigidity mediate the awareness-to-practice transition, this gap is not merely a measurement artefact but a substantive theoretical insight that existing frameworks have yet to fully account for. Empirically, the study provides large-scale quantitative evidence on teacher AI readiness from Sub-Saharan Africa, drawn from a nationally representative, stratified sample of 2,586 teachers across Nigeria's six geopolitical zones. This sample size substantially exceeds minimum power requirements and enhances the generalisability of findings within the Nigerian context, while offering a reference dataset for future comparative work across the Global South. Unlike prior studies that rely on small convenience samples, this research demonstrates that the awareness–skills gap is a systematic, population-level phenomenon rather than an artefact of localised sampling. Practically, by identifying that gender and career stage, rather than institutional type, are the primary demographic moderators of AI pedagogical skills, the study provides actionable, equity-sensitive evidence for designing professional development frameworks that address structural rather than merely individual barriers to AI integration in Nigerian and comparable Sub-Saharan African education systems.

Recommendations

The findings of this study point to several interconnected recommendations for policy, practice, and future research aimed at strengthening AI readiness among secondary school teachers in Nigeria and comparable contexts. These recommendations are grounded in the empirical evidence generated in this study and aligned with emerging international frameworks on AI competence for educators [14,66,68].

First, the finding that Nigerian teachers demonstrate moderate AI awareness but comparatively low pedagogical skills, a gap confirmed across a nationally representative sample of 2,586 teachers, points directly to the need for structured, longitudinal professional development programmes that move beyond introductory AI awareness sessions toward sustained pedagogical capacity building. Awareness-raising alone is demonstrably insufficient; what teachers need are guided, practice-oriented opportunities to design, implement, and critically evaluate AI-enhanced instructional practices within their specific subject and classroom contexts. The moderate awareness scores and comparatively lower pedagogical skill ratings observed in this study indicate that teachers may understand AI at a conceptual level but require guided opportunities to design, implement, and critically evaluate AI-enhanced instructional practices. Professional development should therefore be practice-oriented, incorporating lesson design workshops, peer collaboration, classroom-based experimentation, and reflective evaluation cycles. Evidence from AI-supported instructional research suggests that experiential and simulation-based approaches can enhance teachers' responsiveness and adaptive expertise [5,41]. Embedding such approaches within national teacher development frameworks would support a transition from surface-level adoption to meaningful pedagogical integration.

Second, professional learning initiatives must explicitly integrate ethical reasoning, algorithmic awareness, and inclusive education principles into AI training. AI competence is not solely technical; it encompasses the ability to evaluate bias, ensure data security, and promote equitable learning opportunities. The observed gender differences in AI awareness and pedagogical skills, although small in magnitude, underscore the importance of designing inclusive training environments that mitigate AI anxiety, strengthen confidence, and avoid reinforcing existing digital divides. Research on AI-related anxiety and trust highlights the role of psychological and socio-cultural factors in shaping engagement with AI technologies [21,23]. Consequently, capacity-building initiatives should incorporate mentoring structures, collaborative communities of practice, and gender-sensitive facilitation strategies to ensure equitable participation and outcomes.

Third, the finding that experienced teachers, who constitute 54% of this sample and the majority of Nigeria's secondary school workforce, report significantly lower AI pedagogical skills than their early-career and mid-career colleagues has direct policy implications. Differentiated professional development pathways must be developed to account for this variation across career stages. A one-size-fits-all training model will disproportionately fail the largest segment of the teaching workforce. Experienced teachers specifically may benefit from re-skilling programmes that explicitly connect their established pedagogical expertise with AI-enhanced instructional strategies, validating their professional knowledge while supporting technological renewal, rather than positioning AI adoption as a wholesale replacement of existing practice. Rather than assuming uniform training needs, policymakers and school leaders should adopt tiered intervention models. Experienced teachers may benefit from re-skilling programmes that explicitly connect established pedagogical expertise with AI-enhanced instructional strategies, thereby valuing professional experience while supporting technological renewal. Continuous professional renewal is particularly important in rapidly evolving technological ecosystems, where pedagogical practices can plateau without sustained innovation [26].

Fourth, AI integration strategies should be embedded within broader institutional reform rather than treated as isolated technological initiatives. Although school type did not significantly predict AI awareness or pedagogical skills in this study, systemic support structures remain critical. Education authorities should ensure access to reliable digital infrastructure, curated AI tool repositories aligned with curriculum standards, and technical support systems that reduce the cognitive load associated with experimentation. Integrating AI competence standards into teacher evaluation frameworks and accreditation systems may further institutionalize expectations for AI-informed pedagogy while maintaining flexibility for contextual adaptation.

Fifth, teacher education programmes at the pre-service level should incorporate foundational AI literacy and pedagogical AI modules within their curricula. Preparing future teachers with structured exposure to AI-integrated instructional design, assessment analytics, and ethical considerations will reduce the remediation burden at the in-service stage. Aligning teacher preparation curricula with recognized AI competence frameworks will create a more coherent pipeline of AI-ready educators entering the profession. Recent studies on pre-service teachers' AI readiness and literacy [39,52,53] provide valuable insights for curriculum development in this area.

Finally, further research is recommended to extend and deepen the present findings. Longitudinal studies are needed to examine how AI awareness and pedagogical skills evolve over time and in response to targeted interventions. Mixed-methods research could provide richer insight into the contextual and motivational mechanisms underlying gender and experience differences. Experimental and quasi-experimental designs would be particularly valuable in assessing the causal impact of structured AI professional development programmes on classroom practices and student outcomes. Additionally, cross-national comparative studies across Sub-Saharan Africa would help situate Nigerian findings within a broader regional perspective [32,50], thereby contributing to a more inclusive global evidence base in AI and education research. Investigations into AI anxiety [47,48], AI literacy development [33,40], and AI integration across diverse educational contexts [36,34,46] will further enrich the field.

Collectively, these recommendations emphasize that advancing AI integration in education requires systemic, inclusive, and pedagogically grounded strategies. Moving from awareness to transformative practice demands sustained, coordinated investment in teacher competence, ethical governance, and research-informed professional development. The present findings underscore that teachers must remain central actors in shaping how AI technologies are interpreted, adapted, and enacted within authentic classroom contexts, a principle that should anchor policy design for scalable and equitable AI integration in education.

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Declarations

Code availability

Not applicable.

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CRediT authorship contribution statement

Oluwakemi Dessy Olurinola: Writing – review & editing, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Olugbenga Adeyanju Akintola:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Damola Olugbade:** Writing – review & editing, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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